

Geometry of Particle Methods

Definition (Wasserstein Gradient)

The Wasserstein Gradient of $\mathcal{F} : \mathcal{P}(\mathbb{R}^d) \rightarrow \mathbb{R}$ at the point $\mu \in \mathcal{P}_2(\mathcal{M})$ is (when it exists and is square integrable) the function,

$$\nabla^W \mathcal{F}[\mu](x) = \nabla_x \delta \mathcal{F}[\mu](x).$$

Intuition: the “transport map of steepest descent,”

$$X_{t+\epsilon} \approx X_t + \epsilon \nabla^W \mathcal{F}[\mu](X_t) \quad \mu_{t+\epsilon} \approx (\text{id} + \epsilon \nabla^W \mathcal{F}[\mu])_{\#} \mu_t$$

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Vanilla mean-field WGF: fix μ_0 , sample $X_0^{(1)}, X_0^{(2)}, \dots, X_0^{(n)} \sim \mu_0$ independently, run the dynamics:

$$\begin{cases} \partial_t X_t^{(1)} = \nabla \delta \mathcal{F}[\hat{\mu}_t](X_t^{(1)}) \\ \partial_t X_t^{(2)} = \nabla \delta \mathcal{F}[\hat{\mu}_t](X_t^{(2)}) \\ \vdots \\ \partial_t X_t^{(n)} = \nabla \delta \mathcal{F}[\hat{\mu}_t](X_t^{(n)}) \end{cases}$$

$$\hat{\mu}_t = \frac{1}{n} \sum_{i=1}^n \delta_{X_t^{(i)}}$$

with $X_{t=0}^{(i)} = X_0^{(i)}$, $i = 1 \dots n$.